1. Identity

Document	STAR-PKG-STAR-CALENDAR-EN · v2.2 · 2026-11-02
Package	star-calendar
Version	0.1.0
Capability	star.time.calendar
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_calendar

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_calendar-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_calendar-0.1.0-py3-none-any.whl	1fe3f93786637e524a20c9314b3c5a7255 4d8d010dc24d4fa198f2f737845739
star_calendar.py	5f13630e7f6b18a15f81f71232bdde3fb7 05581e12123545f47f30302d4d7e29
star_calendar-0.1.0-source.zip	e6b2b11ccd6bd78c7478ecfef71564c1e1 84dab5af9403a9840c97adee8bfdff

5. What it does, and its limits

Exact integer conversion between calendar dates and Julian Day Numbers in the Julian, proleptic Gregorian and civil calendars from JDN 0 to year 9999; identical to ERFA, CPython datetime and NAIF SPICE on 9014 dates, and dates that do not exist are refused.

A calendar, not a time scale: no leap seconds, no UTC/TT/TDB conversion, no time zones, no local adoption dates of the Gregorian calendar (the civil switch is the papal one of October 1582).

- `jdn(year, month, day, calendar)` and `calendar_date(jdn, calendar)` for `"gregorian"`, `"julian"` and `"auto"` (Julian up to 1582-10-04, Gregorian from 1582-10-15); astronomical year numbering; valid from JDN 0 to 9999-12-31.
- `julian_date(...)` returns the pair (JD at the preceding midnight, fraction of day): the sum is the Julian Date and the pair keeps a nanosecond that a single float would lose. `weekday(jdn)` (0 = Monday), `day_of_year(...)`.
- Measured: the same day number as ERFA, CPython `datetime` and NAIF SPICE (MIXED calendar) on 9014 dates, zero mismatches, including years B.C. and every day around the 1582 switch.
- Every function returns integers (or the documented pair) or raises `ValueError`: non-integer or boolean year, month, day or day number; a date that does not exist in the calendar (29 February 1900 Gregorian, 1582-10-05 to 1582-10-14 civil); an unknown calendar; a time outside the day; anything outside the valid range.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before

using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_calendar_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-calendar}: Calendar dates and Julian day numbers, exactly},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_calendar/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts